

20.37%

Overall Churn Rate
1 in 5 customers exited

24.98%

High-Value Churn Ratio
Premium customer attrition

32.44%

Geographic Risk Index
Highest regional exposure

26.85%

Engagement Drop Indicator
Inactive customer churn risk

PROJECT OVERVIEW

This project analyses customer behavior and churn patterns across European banking markets using a cleaned dataset of customer demographic, financial, and behavioral attributes. An interactive Streamlit dashboard was built to help banking analysts identify at-risk segments, monitor KPIs in real time, and design targeted retention strategies.

PROBLEM STATEMENT

- Banks lacked visibility into which customer groups are most likely to churn.
- Churn behavior varied by region, age, and gender but was not being quantified.
- Financial impact of losing high-value customers was not measured or monitored.

METHODOLOGY & TECH STACK

- Data Loading:** CSV import via Pandas; cached with Streamlit @st.cache_data
- Segmentation:** Binning on Age, Credit Score, Tenure, Balance & Geography columns
- KPI Engine:** Grouped aggregations using df.groupby() and conditional filtering
- Visualizations:** Interactive bar, line & scatter plots built with Plotly Express
- Dashboard:** Streamlit wide-layout with real-time sidebar filters (5 dimensions)
- Drill-Down:** Churned / Retained / All customer view with dynamic table refresh

Tools: Python • Pandas • NumPy • Plotly • Streamlit • VS Code

REGIONAL & GENDER CHURN BREAKDOWN

Region	Female Churn	Male Churn
France	20.34%	12.71%
Germany	37.55% 	27.81%
Spain	21.21%	13.11%

Germany is the highest-risk market — female customers there churn at nearly 3× the rate of male customers in France. Female customers show higher churn across all three regions, pointing to a service-quality or engagement gap that demands gender-aware retention strategies.

AGE GROUP CHURN FINDINGS

- 45–60 yrs (51%):** Highest churn — middle-aged customers have elevated financial expectations and are most sensitive to competitive offers.
- 60+ yrs (25%):** Moderate risk — seniors may struggle with digital banking adoption.
- 30–45 yrs (16%):** Relatively stable; balanced financial product engagement.
- <30 yrs (8%):** Lowest churn — digital-native customers with strong platform loyalty.

TENURE VS CHURN

- New (21.14%):** Highest risk — onboarding experience and early engagement are critical.
- Long-Term (20.45%):** Loyalty alone does not prevent churn; personalized offers still needed.
- Mid-Term (19.64%):** Most stable group — trust established, products actively used.

STRATEGIC RECOMMENDATIONS

Germany Priority

Immediately escalate region-specific retention campaigns targeting German female customers — the single highest-risk segment (37.55% churn).



Protect High-Value Clients

Launch VIP relationship management programs for customers with balances above 100K; high balance does not prevent churn (24.98% ratio).



Fix Onboarding Gaps

Introduce 90-day onboarding journeys with check-in calls and loyalty rewards to reduce new customer churn (21.14%) in the critical early window.